

HealthFirst Pharmacy — PIMS

Programming 732 Final Project Report

Student: Njabulo Maluleke | Student Number: 402110592

1. Introduction

The HealthFirst Pharmacy Pharmacy Inventory Management System (PIMS) is a Java Swing desktop application designed for a pharmacy scenario. The system manages medicines and suppliers, supports separate Administrator and Cashier roles, processes sales, updates stock and provides business reports. The corrected implementation uses JDBC with MySQL as the system of record. Swing provides the desktop graphical interface, while JDBC provides access to the relational database [1][2].

2. Requirements and Scope

The system implements database-backed authentication, Administrator functions for medicines, suppliers and users, a Cashier point-of-sale workflow, billing, stock validation and four reports. The required database consists of users, suppliers, medicines, sales and sale_items. The implementation uses foreign keys and transactional checkout so that a sale and its inventory changes are committed together.

3. System Design and Implementation

Authentication queries the users table and compares the entered password with its SHA-256 hash. The user's role determines the dashboard that is displayed. Administrator screens query and insert database records for medicines, suppliers and users. The Cashier POS loads available medicines from MySQL, validates quantities against current stock and creates a sale plus normalized sale-item rows at checkout. The stock update is performed in the same database transaction as the sale; failed transactions are rolled back.

4. Database Design

The users table stores login information, role and full name. Suppliers store contact details. Medicines store product information, price, stock, reorder level, expiry date and supplier relationship. Sales store the sale header and cashier. Sale items store each medicine sold, quantity and price at the time of sale. MySQL foreign keys enforce relationships and constraints protect quantities and monetary values [3].

5. User Manual

1) Create the pims database by running database.sql. 2) Ensure MySQL is running. 3) Configure PIMS_DB_URL, PIMS_DB_USER and PIMS_DB_PASSWORD if the local MySQL settings differ. 4) Build with Maven using mvn clean package. 5) Start target/pims-1.0.0.jar. The seeded database accounts are admin/admin123 and cashier/cash123. Administrator users can create additional accounts from the Users tab. Cashiers select medicines, add quantities to the cart, check stock and use Checkout to create a real sale and bill.

6. Testing and Evidence

The corrected Java source was compiled successfully with JDK 21. The previous implementation contained a demonstration fallback, hard-coded authentication and fabricated report rows; these have been removed. The corrected version requires the MySQL database at runtime. For academic evidence, screenshots must be captured from this corrected build after connecting it to MySQL. Earlier screenshots generated by a separate ScreenshotGenerator are not claimed as database-backed evidence and that generator has been removed.

7. GitHub and Version Control

The intended public repository is PROGRAMMING732-PIMS under the current GitHub account njabulomaluleke21-lang. The GitHub connector could not currently find that repository, so this report does not claim that publication has already occurred. The final repository should contain real commits for the project structure, database, JDBC layer, authentication, inventory, suppliers, users, POS, checkout, reports and documentation. GitHub documentation explains repository creation and pushing commits [4][5].

8. Submission Checklist

Included in the corrected project are the Java source, database.sql, Maven configuration, Windows build script, README, configuration example and supporting documentation. The stale demonstration JAR and ScreenshotGenerator were removed. The remaining environment-dependent submission steps are capturing live screenshots with MySQL and publishing the project to the student's authorized public GitHub repository with an auditable 10+ commit history.

9. Conclusion

The corrected HealthFirst Pharmacy PIMS now uses a real MySQL database rather than hard-coded logins, in-memory sample data or fabricated reports. Authentication, inventory, suppliers, users, sales and reporting are database-driven. Checkout uses a transaction and stock locking to keep the sale and inventory consistent. The project is structured for reproducible Maven builds and for publication in the student's current GitHub account. Final live screenshots and GitHub publication should only be claimed after those steps have actually been completed.

References

[1] Oracle. Java Swing Tutorial. <https://docs.oracle.com/javase/tutorial/uiswing/>

[2] MySQL. Connector/J Developer Guide. <https://dev.mysql.com/doc/connector-j/en/>

[3] MySQL. FOREIGN KEY Constraints. <https://dev.mysql.com/doc/refman/8.4/en/create-table-foreign-keys.html>

[4] GitHub Docs. Creating a new repository.
<https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-new-repository>

[5] GitHub Docs. Pushing commits to a remote repository.
<https://docs.github.com/en/get-started/using-git/pushing-commits-to-a-remote-repository>